

Extra Linux Tools Cheat Sheet

Useful tools that usually aren't installed by default: what each one does, and how to install them.

System monitoring

Command	What it does	Command	What it does
htop	Interactive process list (a friendlier top)	btop	CPU, memory, disk, net and processes at once
atop	Like top, plus history you can replay later	glances	One overview of everything, incl. containers
iotop	Which process is using the disk	nmon	Compact performance dashboard; can record
sar	History: "what happened at 3 AM?" (sysstat)	iostat	Disk throughput and wait times (sysstat)

Disk and files

Command	What it does	Command	What it does
ncdu	Interactive "what's using my disk?" browser	gdu	Like ncdu, much faster on SSDs
duf	A friendlier df, in colored tables	mc	Midnight Commander: two-pane file manager
ranger	Lightweight file manager with previews	tree	Shows a folder as a tree
fd	A simpler, faster find (fd '\.log\$')	rg	ripgrep: a much faster grep -r
bat	cat with syntax colors and line numbers	fzf	Fuzzy finder; supercharges Ctrl-R history

Networking

Command	What it does	Command	What it does
mtr	traceroute + ping, updating live	iftop	Live bandwidth per connection
nethogs	Live bandwidth per process	nload	Live in/out traffic graph per interface
iperf3	Measures real speed between two machines	nmap	Finds open ports (scan only your own hosts)
tcpdump	Captures packets to see what's on the wire	ipcalc	Subnet range, netmask, broadcast
dig	DNS lookups (missing on minimal installs)	ethtool	NIC link speed, duplex and settings

Logs, text and workflow

Command	What it does	Command	What it does
lnav	Log viewer: merges, colors, filters logs	jq	Slices and filters JSON
multitail	Watches several logs at once in split panes	tldr	Short, example-first man pages (tldr tar)
mosh	SSH that survives Wi-Fi changes and sleep	pv	Progress bar for any pipe
entr	Reruns a command when files change	screen	Older tmux alternative; keeps sessions alive

Hardware and troubleshooting

Command	What it does	Command	What it does
smartctl	Drive health, before it fails (smartmontools)	sensors	CPU / board temperatures and fans
strace	Shows system calls: "why does it hang?"	lsof	Open files and ports, by process
inxi	One-command hardware summary		

Installing them

System	Install command
Debian, Ubuntu	sudo apt install btop ncdu mtr
Fedora, RHEL, Rocky, Alma	sudo dnf install btop ncdu mtr (RHEL family: run sudo dnf install epel-release first)
Arch · openSUSE · Alpine	sudo pacman -S ... · sudo zypper install ... · sudo apk add ...

Find a package: **apt search ncdu · dnf search ncdu · dnf provides '*/sensors'** (which package has a command). Package names that differ from the command:

Command	Package (Debian/Ubuntu · Fedora/RHEL)	Command	Package (all)
dig	dnsutils · bind-utils	sar iostat	sysstat
sensors	lm-sensors · lm_sensors	smartctl	smartmontools
fd, bat	fd-find, bat (Debian runs them as fdfind, batcat)	rg	ripgrep

Starter set: **sudo apt install btop ncdu mtr iftop ripgrep jq lnav smartmontools tldr**